

Urjit Kurulkar

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TECHNICAL SKILLS

Data Engineering & SQL | SQL, PL/SQL, Python, PySpark, AWS Redshift, Data Pipelines, Data Transformation, Data Modelling, Batch Processing, Kafka (MSK)

Orchestration & Engineering | Airflow, CI/CD, Git, GitHub, Bitbucket, Docker, Unix Shell, Modular Development, Production Monitoring

Cloud & Data Platforms | AWS (S3, Redshift, SageMaker, Bedrock), Cloud Data Warehousing, Large-Scale Data Processing

Data Quality & Reliability | Data Validation, Automated Monitoring, Production Testing, Population Stability Monitoring, Model Validation, Pipeline Reliability

Analytics & Machine Learning | Regression, Classification, Time Series Forecasting, Segmentation, Recommender Systems, NLP, Statistical Testing

GenAI | LLMs, RAG, Embeddings, Vector Databases, Semantic Retrieval, Re-ranking, Agentic AI
BI & Visualisation | Power BI, Tableau

WORK EXPERIENCE

Senior Data Scientist (AI/ML Engineering)

University of Phoenix

Sep 2022 – Present

Arizona, USA

- Developed predictive classification models for student risk, improving advisor decision accuracy by 15% and identification coverage by 25%.
- Designed ML evaluation frameworks covering model validation, backtesting, performance metrics, population stability, and production degradation.
- Conducted deep-dive analyses across behavioral, engagement, and academic data to assess model performance and translate findings into actionable recommendations.
- Built production RAG solutions using LLMs, semantic retrieval, and re-ranking, improving response relevance by 15% across 1,500+ monthly interactions.
- Built an LLM-powered decision-support system generating contextual next-best actions for advisors across 1,500+ monthly interactions.
- Built automated monitoring and reporting workflows to track model performance, data quality, and population shifts across production ML solutions.
- Built reusable ML foundations across 8+ production solutions, reducing onboarding time by 40% and saving 100+ engineering hours annually.
- Developed reusable ML engineering frameworks and Git-based templates for model development, evaluation, containerization, and deployment.
- Standardized inference, monitoring, and retraining workflows, accelerating adoption of production ML practices across teams.
- Built production ML pipelines using Python, SQL, SageMaker, Redshift, Airflow, Kafka, Docker, and CI/CD, supporting models from development through production.
- Partnered with product, engineering, and business teams to define analytical priorities, evaluate model outcomes, and translate findings into product and operational decisions.

Tech Lead / Software Engineer, Telecom Client

Tech Mahindra

Dec 2017 – Mar 2021

Pune, India

- Led development of scalable systems, improving operational efficiency by 600% and reducing processing time by 6+ hours daily
- Built and owned a mission-critical agent commission calculation platform end-to-end (PL/SQL, Java, full-stack), leading its migration from a legacy mainframe system to a modern Java-based architecture
- Delivered multiple 6-7 month project cycles introducing business logic changes, UI enhancements, and database improvements to the commission platform with zero disruption to live operations

- Designed and delivered a production pricing platform, translating complex business logic into scalable analytics and automated decision-support workflows for regional operations
- Built a time-series based predictive model (multi-linear regression) achieving 86.4% accuracy to support demand estimation and strategic pricing decisions
- Automated data processing pipelines using PL/SQL and scripting, reducing turnaround time by ~20% and improving data availability for analytics workflows
- Partnered with client stakeholders on requirements, solution design, demos, and technical delivery

Data Science Intern

Jan 2017 – Jun 2017

Indian Space Research Organisation (ISRO)

Ahmedabad, India

- Enhanced large-scale document indexing and retrieval using Hadoop Distributed File System (HDFS), enabling efficient search across high-volume scientific repositories
- Developed an NLP-powered semantic search system for large-scale document retrieval, improving accessibility of unstructured scientific data

PROJECTS

Netflix Movie Recommendation Engine

Personal Project

- Built an end-to-end AI recommendation platform combining classical recommenders, FAISS semantic retrieval, RAG and LLM-based reranking, deployed through an interactive Streamlit application.
- Implemented LLM-powered conversational recommendations using retrieved user/movie context and evaluated recommendation quality using Precision@K and Recall@K.

Mortgage Default Prediction

Academic Project — University of Illinois at Chicago

- Built Neural Network, Logistic Regression, and Decision Tree models to predict customer loan default; evaluated models using variable importance, Lift, ROC, and Gain charts, selecting the best model by Area Under the Curve

EDUCATION

M.S. Business Analytics

May 2023

University of Illinois at Chicago (GPA – 4.0)

USA

Bachelor of Technology in Computer Science

September 2017

NIIT University

India